

FYP Proposal Submission Form

Revision 1

Detail of Project Members:	ABDUL WASAY AHMAD ALI
Supervised By:	Dr. Mamuna Fatima
Title:	AutoFix: Intelligent Multi-Branch Vehicle Service Hub
Which real world Problem shall be solved by this project?	Multi-branch vehicle service centers struggle with inefficient booking systems, leading to customer frustration and lost revenue. Customers face unclear service pricing, lack real-time updates, and receive no personalized maintenance guidance. Managers cannot effectively track mechanic workloads or monitor inventory levels across branches, causing service delays. Poor communication between branches results in inconsistent service quality and missed business opportunities. Traditional manual processes create data silos, preventing analytical insights for business optimization. These inefficiencies reduce customer satisfaction, increase operational costs, and limit the growth potential of service centers.

Project Description (What, How and Objectives):	Overall working/summary of this FYP: AutoServX is a web-based platform for multi-branch vehicle service centers to manage bookings, services, and spare parts efficiently. Customers can book appointments in real-time, view service durations and prices, interact with an AI assistant for guidance, and provide feedback. Managers can oversee branch operations, track mechanic workload, manage inventories, and generate analytical reports. Integrated notifications and secure payment gateways enhance transparency and convenience. By centralizing all operations, AutoServX improves customer experience, ensures proper inventory management, and provides predictive insights for service optimization. Name and detail of each module in FYP: Module 1: User Authentication & Profile Management Secure JWT-based login/signup with role-based access for Customer, Manager, and Admin. Includes profile management for multiple vehicles. Module 1: Service Booking & Appointment System Real-time booking with calendar integration, slot availability checking, branch working hours validation, and dashboard for service overview. Module 2: Multi-Branch Service & Inventory Management Manages branch profiles, staff allocation, services offered, spare parts tracking across branches, and automated low-stock notifications. Module 3: AI Customer Assistant & Recommendation Engine Intelligent chatbot for customer queries, predictive pricing estimation, service duration forecasting, and seasonal maintenance recommendations. Module 4: Vehicle Service Tracking & Allocation Service history logs, quality tracking with customer feedback, mechanic workload management, and task allocation system. Module 5: Centralized Analytics & Multi-Dashboard System Real-time analytics for bookings, inventory, and revenue with branch-wise KPI dashboards, data visualization, and exportable reports. Module 6: Notification & Payment Integration System Integrated secure payment gateway supporting Credit/Debit cards and Stripe, plus real-time notifications for bookings and updates. Module 7: Customer Feedback & Blogs Rating and review collection system, testimonials management, and educational blogs for car care tips and service awareness. Member-wise Module Information: 1: Module developed by ABDUL WASAY (CIIT/FA22-BCS-127/WAH)? Module 1 : Service Booking & Appointment System Module 3 : AI Customer Assistant & Recommendation Engine Module 5: Centralized Analytics & Multi-Dashboard System Module 7: Customer Feedback & Blogs 2: Module developed by AHMAD ALI (CIIT/FA22-BCS-144/WAH)? Default Module : User Authentication & Profile Management Module 2: Multi-Branch Service & Inventory Management Module 4: Vehicle Service Tracking & Allocation Module 6: Notification & Payment Integration System Similar FYP from RMS student console Spare Parts E-commerce App Three (3) new, but main, functionalities adding to this FYP: AI-powered Customer Assistance & Predictive Recommendations Centralized Analytics Dashboards with Data Visualization Integrated Multi-Branch Inventory Management with Alerts
Project Streams:	